Numerical Methods

Lecture 5-7 Linear Algebra (Matrix)

1. 考试重点

和之前的内容不一样，矩阵部分不光要求能写代码还要求能用公式手写出答案。 题目对于矩阵和线下方程求解有深刻的要求，务必理解下面的全部内容

1.1 矩阵的基本运算：

包括矩阵的乘法、矩阵的加法、矩阵的转置等。

1.2 矩阵的性质：

- 矩阵的行列式：用于判断矩阵是否可逆，以及线性方程组是否有唯一解。
- 矩阵的逆：通过行变换将矩阵化为单位矩阵的同时，单位矩阵会变为原矩阵的逆矩阵。
- 线性方程组的解的性质：包括线性方程组有唯一解、无解和有无穷多解三种情况。

1.3 线性方程组的解法：

- 高斯消元法：通过行变换将线性方程组的增广矩阵化为上三角形式，然后利用回代求解。
- LU分解法：将矩阵分解为一个下三角矩阵和一个上三角矩阵的乘积，然后利用LU分解求解线性方程组。
- 迭代法：如高斯-赛德尔迭代法，通过迭代的方式逐步逼近线性方程组的解。

2. Basic Matrix Operations and Properties（矩阵的基本运算和性质）

2.1 Basic Matrix Operations（矩阵的基本运算）

2.1.1 Matrix Addition and Subtraction（矩阵的加法和减法）

Matrix addition and subtraction are performed element-wise. Given two matrices A and B of the same dimensions, the sum $C = A + B$ and the difference $D = A - B$ are defined as:

$$C_{ij} = A_{ij} + B_{ij}, \quad D_{ij} = A_{ij} - B_{ij}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \quad A - B = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

2.1.2 Matrix Multiplication（矩阵的乘法）

Matrix multiplication is defined for matrices A of size $m \times n$ and B of size $n \times p$. The product $C = AB$ is a matrix of size $m \times p$ with elements:

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

2.1.3 Matrix Transpose（矩阵的转置）

The transpose of a matrix A, denoted A^T, is obtained by swapping the rows and columns of A. If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix with elements:

$$(A^T)_{ij} = A_{ji}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

2.2 Matrix Properties（矩阵的性质）

2.2.1 Determinant（行列式）

The determinant of a square matrix A, denoted $\det(A)$ or $|A|$, is a scalar value that can be computed from the elements of A. It is used to determine if the matrix is invertible（可逆）. If $\det(A) \neq 0$, the matrix is invertible; otherwise, it is not.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2$$

Since $\det(A) \neq 0$, A is invertible.

2.2.2 Inverse（逆矩阵）

The inverse of a square matrix A, denoted A^{-1}, is a matrix such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix（单位矩阵）. The inverse exists if and only if $\det(A) \neq 0$.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$$

2.2.3 Solutions of Linear Systems（线性方程组的解）

A linear system $A\boldsymbol{x} = \boldsymbol{b}$ can have:

- **Unique Solution**（唯一解）: If $\det(A) \neq 0$, the system has a unique solution.
- **No Solution**（无解）: If the system is inconsistent.
- **Infinitely Many Solutions**（无穷多解）: If the system is dependent.

3. Calculation（计算方法）

3.1 Gaussian Elimination（高斯消元法）

3.1.1 Overview

Gaussian Elimination is a method for solving systems of linear equations by transforming the augmented matrix（增广矩阵）into an upper-triangular form（上三角形式）. This process involves a series of row operations（行操作）to simplify the matrix, making it easier to solve for the unknowns using back substitution（回代）.

3.1.2 Step-by-Step Process

Consider the following system of linear equations:

$$\begin{align} x + 2y + 4z &= -20 \tag{1} \\ 2x + 5y + 6z &= -36 \tag{2} \\ -x - 5y + 3z &= 26 \tag{3} \end{align}$$

3.1.2.1 Write the Augmented Matrix（写出增广矩阵）

First, we write the augmented matrix for the system:

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & -20 \\ 2 & 5 & 6 & -36 \\ -1 & -5 & 3 & 26 \end{array}\right]$$

3.1.2.2 Eliminate the First Column Below the Pivot（消去第一列的主元下方元素）

The goal is to make the elements below the first pivot（主元）equal to zero. We use row operations to achieve this:

- Subtract 2 times the first row from the second row:

$$R2 \leftarrow R2 - 2R1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & -20 \\ 0 & 1 & -2 & 4 \\ -1 & -5 & 3 & 26 \end{array}\right]$$

- Add the first row to the third row:

$$R3 \leftarrow R3 + R1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & -20 \\ 0 & 1 & -2 & 4 \\ 0 & -3 & 7 & 6 \end{array}\right]$$

Explanation: The purpose of these operations is to create zeros below the first pivot (1 in the first row). This simplifies the matrix and brings it closer to upper-triangular form.

3.1.2.3 Eliminate the Second Column Below the Pivot（消去第二列的主元下方元素）

Next, we eliminate the elements below the second pivot:

- Add 3 times the second row to the third row:

$$R3 \leftarrow R3 + 3R2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & -20 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & 1 & 18 \end{array}\right]$$

Explanation: By adding a multiple of the second row to the third row, we create a zero below the second pivot (1 in the second row). This step further simplifies the matrix and brings it closer to upper-triangular form.

3.1.2.4 Back Substitution（回代求解）

Now that the matrix is in upper-triangular form, we can solve for the unknowns using back substitution:

- Solve for z:

$$z = 18$$

- Solve for y:

$$y - 2z = 4 \implies y - 2(18) = 4 \implies y = 40$$

- Solve for x:

$$x + 2y + 4z = -20 \implies x + 2(40) + 4(18) = -20 \implies x = -172$$

Explanation: Back substitution starts from the bottom row and works its way up. Since the matrix is upper-triangular, each equation has only one unknown variable that has not been solved yet, making it easy to find the values step by step.

3.1.2.5 Summary

The solution to the system is:

$$\boxed{x = -172, y = 40, z = 18}$$

3.2 LU Decomposition（LU分解法）

3.2.1 Overview

LU Decomposition is a technique that decomposes a matrix A into a lower-triangular matrix L（下三角矩阵）and an upper-triangular matrix U（上三角矩阵）such that $A = LU$. This decomposition allows us to solve the system $A\boldsymbol{x} = \boldsymbol{b}$ more efficiently by solving two simpler triangular systems.

3.2.2 Step-by-Step Process

Consider the matrix A:

$$A = \begin{bmatrix} 2 & 3 & 1 & 5 \\ 8 & 16 & 7 & 22 \\ 2 & 15 & 13 & 19 \\ -4 & 2 & 10 & 11 \end{bmatrix}$$

3.2.2.1 Perform Row Operations（执行行操作）

We will perform row operations to transform A into an upper-triangular matrix U, while recording the operations to construct the lower-triangular matrix L.

Step 1: Eliminate the first column below the pivot

- Subtract 4 times the first row from the second row:

$$R2 \leftarrow R2 - 4R1$$

- Add the first row to the third row:

$$R3 \leftarrow R3 - R1$$

- Add 2 times the first row to the fourth row:

$$R4 \leftarrow R4 + 2R1$$

Step 2: Eliminate the second column below the pivot

- Subtract 3 times the second row from the third row:

$$R3 \leftarrow R3 - 3R2$$

- Subtract 2 times the second row from the fourth row:

$$R4 \leftarrow R4 - 2R2$$

Step 3: Eliminate the third column below the pivot

- Subtract 2 times the third row from the fourth row:

$$R4 \leftarrow R4 - 2R3$$

3.2.2.2 Construct L and U（构造 L 和 U）

After performing the row operations, we obtain:

$$U = \begin{bmatrix} 2 & 3 & 1 & 5 \\ 0 & 4 & 3 & 2 \\ 0 & 0 & 3 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The lower-triangular matrix L is constructed by recording the multipliers used in the row operations:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ -2 & 2 & 2 & 1 \end{bmatrix}$$

Explanation: The matrix L records the steps we took to transform A into U. Each entry below the diagonal in L corresponds to a multiplier used in the row operations.

3.2.2.3 Solve the System Using L and U（利用 L 和 U 求解线性方程组）

Given a right-hand side vector $\boldsymbol{b}$, we solve the system $A\boldsymbol{x} = \boldsymbol{b}$ by solving two triangular systems:

1. Solve $L\boldsymbol{c} = \boldsymbol{b}$ for $\boldsymbol{c}$ using forward substitution（前代）.
2. Solve $U\boldsymbol{x} = \boldsymbol{c}$ for $\boldsymbol{x}$ using back substitution（回代）.

Example: Let $\boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$.

Step 1: Solve $L\boldsymbol{c} = \boldsymbol{b}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ -2 & 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Step 2: Solve $U\boldsymbol{x} = \boldsymbol{c}$

$$\begin{bmatrix} 2 & 3 & 1 & 5 \\ 0 & 4 & 3 & 2 \\ 0 & 0 & 3 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

3.3 Gauss-Seidel Iteration（高斯-赛德尔迭代法）

3.3.1 Overview

Gauss-Seidel Iteration is an iterative method for solving systems of linear equations. It updates the solution iteratively by using the latest available values for each variable.

3.3.2 Step-by-Step Process

Consider the following system of linear equations:

$$\begin{aligned} x + 2y + 4z &= -20 \quad (4) \\ 2x + 5y + 6z &= -36 \quad (5) \\ -x - 5y + 3z &= 26 \quad (6) \end{aligned}$$

3.3.2.1 Rewrite the Equations（重写方程）

Rewrite each equation to solve for one variable:

$$\begin{aligned} x &= -20 - 2y - 4z \quad (7) \\ y &= \frac{-36 - 2x - 6z}{5} \quad (8) \\ z &= \frac{26 + x + 5y}{3} \quad (9) \end{aligned}$$

3.3.2.2 Initialize the Solution（初始化解）

Start with an initial guess for the solution, such as $\boldsymbol{x}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

3.3.2.3 Iterate（迭代）

Update the solution iteratively:

$$\begin{aligned} x^{(k+1)} &= -20 - 2y^{(k)} - 4z^{(k)} \quad (10) \\ y^{(k+1)} &= \frac{-36 - 2x^{(k+1)} - 6z^{(k)}}{5} \quad (11) \\ z^{(k+1)} &= \frac{26 + x^{(k+1)} + 5y^{(k+1)}}{3} \quad (12) \end{aligned}$$

Example: Perform a few iterations starting from $\boldsymbol{x}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Iteration 1:

$$\begin{aligned} x^{(1)} &= -20 - 2(0) - 4(0) = -20 \quad (13) \\ y^{(1)} &= \frac{-36 - 2(-20) - 6(0)}{5} = \frac{-36 + 40}{5} = 0.8 \quad (14) \\ z^{(1)} &= \frac{26 + (-20) + 5(0.8)}{3} = \frac{26 - 20 + 4}{3} = \frac{10}{3} \approx 3.33 \quad (15) \end{aligned}$$

Iteration 2:

$$x^{(2)} = -20 - 2(0.8) - 4(3.33) = -20 - 1.6 - 13.32 = -34.92$$
$$y^{(2)} = \frac{-36 - 2(-34.92) - 6(3.33)}{5} = \frac{-36 + 69.84 - 19.98}{5} = \frac{13.86}{5} = 2.$$
$$z^{(2)} = \frac{26 + (-34.92) + 5(2.772)}{3} = \frac{26 - 34.92 + 13.86}{3} = \frac{4.94}{3} \approx 1.65$$

Explanation: Each iteration updates the solution using the latest values. The process continues until the solution converges to a stable value.

3.3.2.4 Convergence（收敛性）

Gauss-Seidel Iteration converges if the matrix A is strictly diagonally dominant（严格对角占优）or symmetric and positive definite（对称正定）. In practice, the iteration is stopped when the change in the solution is below a certain tolerance.

3.4 Additional Examples with Jupyter-Compatible Format

Example 1: Solving a System Using Gaussian Elimination

Consider the system:

$$x + 2y - 2z = 7 \tag{19}$$
$$-x + 0y + 8z = 7 \tag{20}$$
$$2x + 6y + 2z = 30 \tag{21}$$

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & 7 \\ -1 & 0 & 8 & 7 \\ 2 & 6 & 2 & 30 \end{array}\right]$$

Step 1: Eliminate first column below pivot

$$R_2 \leftarrow R_2 + R_1$$

$$R_3 \leftarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & 7 \\ 0 & 2 & 6 & 14 \\ 0 & 2 & 6 & 16 \end{array}\right]$$

Step 2: Eliminate second column below pivot

$$R_3 \leftarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & 7 \\ 0 & 2 & 6 & 14 \\ 0 & 0 & 0 & 2 \end{array}\right]$$

Since the last row represents $0 = 2$, this system has **no solution**.

Example 2: Matrix Operations

Given matrices:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 2 \end{bmatrix}$$

Matrix Multiplication AB:

$$AB = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1(2) + 3(0) + 2(1) & 1(1) + 3(3) + 2(2) \\ 4(2) + 0(0) + 1(1) & 4(1) + 0(3) + 1(2) \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 & 14 \\ 9 & 6 \end{bmatrix}$$